Laura Sommovigo

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Research Profile

I am an astrophysicist working on the formation and evolution of galaxies in the first billion years of cosmic history, with a particular focus on the physics of dust. My research combines ALMA and JWST observations with physically motivated modelling and numerical simulations to understand how dust forms, evolves, and reshapes the observable properties of early galaxies. I develop multi-scale frameworks connecting ISM physics to galaxy-scale observables, with particular emphasis on radiative transfer, synthetic observations, and forward modelling of galaxies at cosmic dawn. I am co-PI and theory lead of the ALMA Large Program PHOENIX and a co-investigator on over 30 successful ALMA, JWST, and Roman proposals, including the ALMA Large Programs HIDING and CHAMPS.

Career Stage & Metrics

PhD awarded	2023, Scuola Normale Superiore; 3 years post-PhD.
Publications	68 refereed journal publications, including 8 first-author papers (NASA ADS, accessed September 2026).
Citation metrics	h-index = 32, total refereed citations = 3164, first-author citations = 404 (NASA ADS, accessed September 2026).

Academic Appointments

2025 – Present **Learning The Universe Research Fellow**, Department of Astronomy, Columbia University, 550 W 120th St, New York, NY 10025, USA

2023 – 2025 **Flatiron Research Fellow**, Flatiron Institute, Center for Computational Astrophysics, New York, NY, USA

Education

2018 – 2023 **Ph.D. in Astrophysics**, Scuola Normale Superiore (SNS), Pisa, Italy. (Awarded: 28 July 2023, with distinction)

2016 – 2018 **MSc in Astrophysics**, Pisa University, Pisa, Italy

2013 – 2016 **BSc in Physics**, Pisa University, Pisa, Italy

Experience

Leadership & Scientific Responsibility

2026 ◇ **Organizer**, Lorentz Center Workshop "From Dust till Dawn", Leiden, Netherlands

2026 – Present ◇ **Team Leader**, ISSI International Team 26-690, International Space Science Institute, Bern, Switzerland

2025 – Present ◇ **Co-Principal Investigator**, ALMA Large Program *PHOENIX*, and **Lead** of the theory group

2025 – Present ◇ **External Panelist**, JWST High-Redshift & Distant Universe Proposals; NSF-AAG Galaxy Evolution Panel

2025 ◇ **Scientific Organizing Committee**, "Galaxy Origins in the JWST Era", Toledo, Spain

2024 – Present ◇ **Co-Lead**, Synthetic Observation Working Group, LTU Collaboration

Major Collaborations

2021 – Present ◇ REBELS (ALMA Large Program; P.I. R. Bouwens)

2024 – Present ◇ ALPINE+CRISTAL (ALMA–JWST Large Programs; P.I. R. Herrera-Camus, A. Faisst)

2024 – Present ◇ CHAMPS (ALMA Large Program; P.I. A. Faisst)

2023 – Present ◇ AtLAST Collaboration – WP6 Science (Distant Galaxies)

2025 – Present ◇ HIDING Collaboration (ALMA Large Programs; P.I. L. Boogaard)

Observing Programmes

. 2025 ◇ **Co-PI**, "PHOENIX: the Emergence of Dust, Obscured Star Formation and ISM Physics at Cosmic Dawn", 2025.1.01606.L, ALMA Large Program (PI S. Schouws)

2025 ◇ **Co-I**, “HIDING in the HUDF: High-definition Dust Imaging of Normal Galaxies in the Hubble Ultra Deep Field”, 2025.1.01377.L, ALMA Large Program (PI L. Boogaard)

Experience (continued)

2023 ◇ **Co-I**, "The COSMOS High-z ALMA-MIRI Population Survey (CHAMPS): A Wide-Area Comprehensive Survey of the Dusty Universe", 2023.1.00180.L, ALMA Large Program (PI A. Faisst)

2025 ◇ **Co-I**, "Pushing Little Red Dots to the Early Universe: Spectroscopic Confirmation of the First Robust $z \sim 10$ LRD Candidate", GO 9483, JWST Cycle 5 (PI T. Tanaka)

2023 ◇ **Co-I**, "Revealing the Lifecycle and Environment of Massive $z \sim 7$ Galaxies", GO 6480, JWST Cycle 3 (PI S. Schouws), 47.1 h

2025 ◇ **Co-I**, "R-HIVE: Mapping the SFH-Morphology-Environment connection with Roman", Program 19063, Roman Cycle 1 (Analysis; PI K. Iyer)

2026 ◇ **Co-I**, "Gas in a cosmic centrifuge: Resolved outflows of a $z = 4.4$ spiral galaxy", 2026.1.01554.S, ALMA (PI U. Steinwandel)

2026 ◇ **Co-I**, "Constraining the dust properties of a luminous UV bump galaxy within the Epoch of Reionisation", 2026.1.01430.S, ALMA (PI K. Ormerod)

2026 ◇ **Co-I**, "Efficiently unveiling luminous dust reservoirs in the first billion years", 2026.1.00854.S, ALMA (PI H. Algera)

2026 ◇ **Co-I**, "RESKUE SF: REsolving the Schmidt-Kennicutt relation of Ultra-luminous Early Starbursts down to Faint regions", 2026.1.00811.S, ALMA (PI P. Arriagada Torres)

2025 ◇ **Co-I**, "Resolved dust emission in massive star forming galaxies in the Epoch of Reionization", 2025.1.01318.S, ALMA (PI M. Stefanon)

2025 ◇ **Co-I**, "The Red Giant at Cosmic Dawn: Quenching in the First Billion Years", 2025.1.01062.S, ALMA (PI K. Chworowsky)

2025 ◇ **Co-I**, "Dust at Dawn: a dusty galaxy at redshift 11", 2025.1.00806.S, ALMA (PI H. Algera)

2025 ◇ **Co-I**, "Probing the neutral gas-phase of star-forming galaxies in the epoch of reionization through systematic [O I] observations", 2025.1.00422.S, ALMA (PI K. Heintz)

Experience (continued)

2025 ◇ **Co-I**, "Unveiling the rapid rise of dust at Cosmic Dawn", 2025.1.00386.S, ALMA (PI H. Algera)

2025 ◇ **Co-I**, "Cold gas in the brightest [O III] 88 μm-emitter at $z > 8.5$", 2025.1.00264.S, ALMA (PI H. Algera)

2025 ◇ **Co-I**, "Confirming Three Hidden ULIRGs in the Epoch of Reionization", 2025.1.00232.S, ALMA (PI H. Algera)

2024 ◇ **Co-I**, "Unveiling the ISM at $z > 14$: A Critical Study of [C II]158 μm in GS-z14", 2024.A.00007.S, ALMA DDT (PI S. Schouws)

2024 ◇ **Co-I**, "[O I]146 at $z = 8.31$: unveiling the high-density ISM in the earliest galaxies", 2024.1.00995.S, ALMA (PI R. Smit)

2024 ◇ **Co-I**, "A feverish galaxy at $z > 8$? ALMA band 9 as a cosmic thermometer", 2024.1.00537.S, ALMA (PI J. Molina)

2024 ◇ **Co-I**, "Where is the Dust?", 2024.1.00440.S, ALMA (PI H. Algera)

2024 ◇ **Co-I**, "The Rise of Ziggy Stardust: Dust Build-up in the Epoch of Reionization", 2024.1.00406.S, ALMA (PI H. Algera)

2023 ◇ **Co-I**, "The ISM at Cosmic Dawn: [O III] in an Uniquely Luminous $z = 14.3$ Galaxy", 2023.A.00037.S, ALMA DDT (PI S. Schouws)

2023 ◇ **Co-I**, "Mapping out the Changing Dust Properties of High-redshift Galaxies", 2023.1.01033.S, ALMA (PI H. Algera)

2022 ◇ **Co-I**, "The kpc-scale Dust Temperature, Star Formation and ISM at $z = 7.3$", 2022.1.01324.S, ALMA (PI H. Algera)

2022 ◇ **Co-I**, "Statistically Characterizing the Oldest, Most Massive Galaxies in the Reionization Era", 2022.1.01007.S, ALMA (PI R. Endsley)

2022 ◇ **Co-I**, "Revolutionary insights into the $z > 7$ gas and dust physics", 2022.1.00999.S, ALMA (PI I. De Looze)

Experience (continued)

2021 ◇ **Co-I**, "Sub-kpc mapping of the [C II]/dust in the highest-redshift ULIRG at $z = 7.3$", 2021.1.01603.S, ALMA (PI J. Hodge)

2021 ◇ **Co-I**, "Revolutionary insights into the $z > 7$ gas and dust physics", 2021.1.01495.S, ALMA (PI I. De Looze)

2021 ◇ **Co-I**, "Exploring the ISM conditions of UV-selected galaxies in the Epoch of Reionization by Surveying [O III], [C II], and Dust", 2021.1.00722.S, ALMA (PI S. Schouws)

2021 ◇ **Co-I**, "The golden ratio: [O III]-to-[C II] ratio census in SFR-selected sample", 2021.1.00668.S, ALMA (PI T. Bakx)

2021 ◇ **Co-I**, "Climbing the CO excitation ladder in the early Universe", 2021.1.00453.S, ALMA (PI S. Carniani)

2021 ◇ **Co-I**, "Hot or Warm? What are the dust temperatures of normal star-forming galaxies at $z \sim 7$?", 2021.1.00318.S, ALMA (PI H. Inami)

Supervision & Mentorship

2023 – 2024 ◇ **MSc Co-Supervisor**, P. Ghosh (with A. Ferrara & P. Dayal)

2024 – Present ◇ **Research Project Mentor**, K. Matsumoto (University of Ghent; supervisor: M. Baes)

2025 – Present ◇ **Research Project Mentor**, B. Casavecchia (MPA; supervisor: B. Ciardi)

2025 – Present ◇ **Research Project Mentor**, M. Manju (NYU; supervisor: A. Pullen)

2025 – Present ◇ **Research Project Mentor**, S. Newman (University of Portsmouth; supervisors: C. Lovell & C. Maraston)

Present ◇ **Research Project Mentor**, E. Rusta (University of Florence; supervisor: S. Salvadori)

Honors & Distinctions

2023 ◇ MPA Postdoctoral Fellowship (Declined)

2023 ◇ Ph.D., Cum Laude – Scuola Normale Superiore

Experience (continued)

2018 ◇ MSc, Cum Laude – University of Pisa

Scientific Service

2021 – Present ◇ Referee for MNRAS, A&A, ApJ

Outreach & Diversity

2024 ◇ **Simons-NSBP Scholars Program Mentor**, New York, USA

2021 ◇ **Founder & Organizer**, "ISPIRA" (SNS-funded outreach initiative); organized ∼40 seminars promoting careers in science for teenage girls

2022 ◇ JWST outreach, *EELISAweek*, Pisa, Italy

2021 ◇ Scientific outreach, *GALILEOMOBILE*, Pisa, Italy

2019 ◇ Outreach seminar on Cosmology, "Spazio allo spazio", Italy

Research Publications

Full publication list available on NASA ADS.

First-Author Publications

1. **Sommovigo, L.**, Bartlett, D. J., Cochrane, R. K., Ho, M., Lovell, C. C., & Somerville, R. S. (2026). Learning the Universe: The Structure of Dust Attenuation Curves in Galaxy Simulations. *ApJ*, *1008*(2), 218. doi:10.3847/1538-4357/ae8e78. arXiv: 2606.10027 [astro-ph.GA]
2. **Sommovigo, L.**, Lancaster, L., Menon, S. H., O'Leary, J. A., Somerville, R. S., & Bryan, G. L. (2026). Blue Monsters and Dusty Descendants: Reconciling UV and IR Emission from Galaxies from z ∼ 7, up to z = 14. *ApJ*, *1003*(2), 170. doi:10.3847/1538-4357/ae5e58. arXiv: 2602.18556 [astro-ph.GA]
3. **Sommovigo, L.**, Cochrane, R. K., Somerville, R. S., Hayward, C. C., Lovell, C. C., Starkenburg, T., … Perez, L. A. (2025). Learning the Universe: Physically Motivated Priors for Dust Attenuation Curves. *ApJ*, *990*(2), 114. doi:10.3847/1538-4357/addec1. arXiv: 2502.13240 [astro-ph.GA]
4. **Sommovigo, L.**, & Algera, H. (2025). Realistic multitemperature dust: how well can we constrain the dust properties of high-redshift galaxies? *MNRAS*, *540*(4), 3693–3708. doi:10.1093/mnras/staf897. arXiv: 2505.20105 [astro-ph.GA]

5. **Sommovigo, L.**, Ferrara, A., Carniani, S., Pallottini, A., Dayal, P., Pizzati, E., … Faisst, A. (2022). A new look at the infrared properties of z∼5 galaxies. *MNRAS*, *517*(4), 5930–5941. doi:10.1093/mnras/stac2997. arXiv: 2210.09312 [astro-ph.GA]

6. **Sommovigo, L.**, Ferrara, A., Pallottini, A., Dayal, P., Bouwens, R. J., Smit, R., … Topping, M. (2022). The ALMA REBELS Survey: cosmic dust temperature evolution out to z∼7. *MNRAS*, *513*(3), 3122–3135. doi:10.1093/mnras/stac302. arXiv: 2202.01227 [astro-ph.GA]

7. **Sommovigo, L.**, Ferrara, A., Carniani, S., Zanella, A., Pallottini, A., Gallerani, S., & Vallini, L. (2021). Dust temperature in ALMA [C II]-detected high-z galaxies. *MNRAS*, *503*(4), 4878–4891. doi:10.1093/mnras/stab720. arXiv: 2102.08950 [astro-ph.GA]

8. **Sommovigo, L.**, Ferrara, A., Pallottini, A., Carniani, S., Gallerani, S., & Decataldo, D. (2020). Warm dust in high-z galaxies: origin and implications. *MNRAS*, *497*(1), 956–968. doi:10.1093/mnras/staa1959. arXiv: 2004.09528 [astro-ph.GA]

Co-Author Publications

1. Casey, C. M., Akins, H. B., Battisti, A. J., McKinney, J., Treister, E., Zavala, J. A., … Zimmerman, D. (2026). Dust in the Average Galaxy: Attenuation, Emission, and Opacity from $0 < z < 7$. *The Open Journal of Astrophysics*, *9*, 67549. doi:10.33232/001c.167549. arXiv: 2606.17270 [astro-ph.GA]

2. Algera, H., Weaver, J. R., Bakx, T. J. L. C., Aravena, M., Bouwens, R. J., Cescon, K., … Xiao, M. (2026). A first systematic study of [OIII] 88 μm at $z > 8$: two luminous oxygen lines and a powerful ionized outflow in the first 600 million years. *The Open Journal of Astrophysics*, *9*, 65333. doi:10.33232/001c.165333. arXiv: 2512.14486 [astro-ph.GA]

3. Cescon, K., Hodge, J. A., Boogaard, L. A., Algera, H. S. B., Rowland, L. E., Riechers, D. A., … Vallini, L. (2026). Direct detection of cool molecular gas in a star-forming galaxy at z=7.31. *MNRAS*, *549*(3), stag924. doi:10.1093/mnras/stag924. arXiv: 2606.13393 [astro-ph.GA]

4. Davies, R. L., Fisher, D. B., Herrera-Camus, R., Faisst, A., Spilker, J., González-López, J., … Zamorani, G. (2026). Multiphase images of a powerful supernova-driven wind in the early Universe. *MNRAS*, *549*(3), stag874. doi:10.1093/mnras/stag874. arXiv: 2606.10271 [astro-ph.GA]

5. Faisst, A. L., Liu, L.-J., Dubois, Y., Nakazato, Y., Osman, O., Pallottini, A., … Zamorani, G. (2026). The ALPINE–CRISTAL–JWST Survey: The Fast Metal Enrichment of Massive Galaxies at z ∼ 5. *1004*(1), 22. doi:10.3847/1538-4357/ae6645. arXiv: 2510.16106 [astro-ph.GA]

6. Chworowsky, K., Finkelstein, S. L., Taylor, A. J., Morales, A. M., Dickinson, M., Yung, L. Y. A., … the CAPERS Collaboration. (2026). Massive Galaxies Form Early and Gray: Stellar Assembly and Dust Attenuation at $\mathbf{z > 3.5}$ from CAPERS. *arXiv e-prints*, arXiv:2605.13966. doi:10.48550/arXiv.2605.13966. arXiv: 2605.13966 [astro-ph.GA]

7. Komarova, L., Stefanon, M., Laza-Ramos, A., Algera, H. S. B., Aravena, M., Bouwens, R., … van der Werf, P. (2026). REBELS-IFU: Spatially Resolved Ionizing Photon Production Efficiencies of 12 Bright Galaxies in the Epoch of Reionization. *ApJ*, *1000*(2), 228. doi:10.3847/1538-4357/ae474f. arXiv: 2511.10743 [astro-ph.GA]

8. Endsley, R., Shapley, A. E., Topping, M. W., Stark, D. P., Bouwens, R. J., Rowland, L. E., … Tang, M. (2026). REBELS–MOSFIRE: Weak C III] Emission Is Typical among Extremely UV-bright, Massive Galaxies at z ∼ 7. *ApJ*, *999*(1), 95. doi:10.3847/1538-4357/ae3a7b. arXiv: 2506.21674 [astro-ph.GA]

9. Fisher, R., Bowler, R. A. A., Cochrane, R. K., Rowland, L. E., Stefanon, M., Algera, H. S. B., … van der Werf, P. P. (2026). REBELS-IFU: Steeply rising star formation histories and the importance of dust obscuration in massive z ∼ 7 galaxies revealed by multiwavelength observations. *MNRAS*, *546*(3), stag049. doi:10.1093/mnras/stag049. arXiv: 2511.10741 [astro-ph.GA]

10. Roper, W., Lovell, C., Vijayan, A., Wilkins, S., Akins, H., Berger, S., … Rawlings, A. (2026). Synthesizer: Synthetic Observables for Modern Astronomy. *The Journal of Open Source Software*, *11*(119), 9436. doi:10.21105/joss.09436. arXiv: 2506.15811 [astro-ph.IM]

11. Bakx, T. J. L. C., Algera, H. S. B., Jolly, J.-B., Esmerian, C., Knudsen, K., **Sommovigo**, **L.**, … Vlemmings, W. (2026). Probing infrared eXcess to investigate early-Universe dust (PIXIEDust). *MNRAS*, *546*(2), staf2284. doi:10.1093/mnras/staf2284. arXiv: 2512.07964 [astro-ph.GA]

12. Rowland, L. E., Stefanon, M., Bouwens, R., Hodge, J., Algera, H., Fisher, R., … van der Werf, P. (2026). REBELS-IFU: evidence for metal-rich massive galaxies at z ∼ 6−8. *MNRAS*, *546*(2), staf2023. doi:10.1093/mnras/staf2023. arXiv: 2501.10559 [astro-ph.GA]

13. Tsujita, A., Fujimoto, S., Faisst, A., Boquien, M., Li, J., Ferrara, A., … Alpine+Cristal Collaborations. (2026). The ALPINE-CRISTAL-JWST Survey: Stellar and Nebular Dust Attenuation of Main-sequence Galaxies at z ∼ 46. *ApJ*, *997*(2), 319. doi:10.3847/1538-4357/ae22d8. arXiv: 2510.18248 [astro-ph.GA]

14. Algera, H. S. B., Rowland, L., Stefanon, M., Palla, M., **Sommovigo**, **L.**, Inami, H., … van der Werf, P. (2026). REBELS-IFU: dust build-up in massive galaxies at redshift 7. *MNRAS*, *545*(2), staf1897. doi:10.1093/mnras/staf1897. arXiv: 2501.10508 [astro-ph.GA]

15. Faisst, A. L., Fujimoto, S., Tsujita, A., Wang, W., Nezhad, N., Loiacono, F., … Zamorani, G. (2026). The ALPINE-CRISTAL-JWST Survey: JWST/IFU Optical Observations for 18 Main-sequence Galaxies at z = 4–6. *ApJS*, *282*(1), 19. doi:10.3847/1538-4365/ae0928. arXiv: 2510.16111 [astro-ph.GA]

16. Matsumoto, K., **Sommovigo**, **L.**, Gebek, A., Nagamine, K., Nersesian, A., Baes, M., … Cochrane, R. K. (2026). Evolution of galaxy attenuation curves driven by evolving dust mass and grain size distributions. *A&A*, *705*, A75. doi:10.1051/0004-6361/202555658. arXiv: 2508.21157 [astro-ph.GA]

17 Algera, H. S. B., Herrera-Camus, R., Aravena, M., Assef, R., Bakx, T. L. J. C., Bolatto, A., … van der Werf, P. (2025). How much gas and dust is in the $z = 5.7$ Lyman Break Galaxy HZ10? An ALMA Band 10 to 4 and JWST/NIRSpec study of its interstellar medium. *arXiv e-prints*, arXiv:2512.02320. doi:10.48550/arXiv.2512.02320. arXiv: 2512.02320 [astro-ph.GA]

18 Bakx, T. J. L. C., **Sommovigo**, **L.**, Tamura, Y., Smit, R., Ferrara, A., Algera, H., … Yoshida, N. (2025). A warm ultraluminous infrared galaxy just 600 million years after the big bang. *MNRAS*, *544*(2), 1502–1513. doi:10.1093/mnras/staf1714. arXiv: 2511.08327 [astro-ph.GA]

19 Iyer, K. G., Starkenburg, T. K., Bryan, G. L., Somerville, R. S., Alfonzo, J. P., Anglés-Alcázar, D., … Wu, J. F. (2025). How Does Feedback Affect the Star Formation Histories of Galaxies? *ApJ*, *994*(2), 174. doi:10.3847/1538-4357/ae0334. arXiv: 2508.21152 [astro-ph.GA]

20 Knudsen, K. K., Richard, J., Jauzac, M., Bakx, T. J. L. C., Goncalves, T. S., Egami, E., … Stark, D. P. (2025). Highly-ionized gas in lensed z = 6.027 Little Red Dot seen through [OIII] 88micron with ALMA. *arXiv e-prints*, arXiv:2512.05097. doi:10.48550/arXiv.2512.05097. arXiv: 2512.05097 [astro-ph.GA]

21 Lovell, C. C., Starkenburg, T., Ho, M., Anglés-Alcázar, D., Davé, R., Gabrielpillai, A., … Villaescusa-Navarro, F. (2025). Learning the Universe: cosmological and astrophysical parameter inference with galaxy luminosity functions and colours. *MNRAS*, *544*(4), 3949–3979. doi:10.1093/mnras/staf1888. arXiv: 2411.13960 [astro-ph.GA]

22 Somerville, R. S., Yung, L. Y. A., Lancaster, L., Menon, S., **Sommovigo**, **L.**, & Finkelstein, S. L. (2025). Density-modulated star formation efficiency: implications for the observed abundance of ultraviolet luminous galaxies at z > 10. *MNRAS*, *544*(4), 3774–3798. doi:10.1093/mnras/staf1824. arXiv: 2505.05442 [astro-ph.GA]

23 Tanaka, T. S., Akins, H. B., Harikane, Y., Silverman, J. D., Casey, C. M., Inayoshi, K., … Sanders, D. B. (2025). Discovery of a Little Red Dot Candidate at z > 10 in COSMOS-web Based on MIRI-NIRCam Selection. *ApJ*, *995*(1), 21. doi:10.3847/1538-4357/ae145f. arXiv: 2508.00057 [astro-ph.GA]

24 Bennett, J. S., Smith, M. C., Fielding, D. B., Bryan, G. L., Kim, C.-G., Springel, V., … **Sommovigo**, **L.** (2025). Prevention is better than cure? Feedback from high specific energy winds in cosmological simulations with ARKENSTONE. *MNRAS*, *543*(2), 1456–1478. doi:10.1093/mnras/staf1440. arXiv: 2410.12909 [astro-ph.GA]

25 Algera, H., Rowland, L., Smit, R., Fisher, R., Ramambason, L., Kumari, N., … Witstok, J. (2025). REBELS-IFU: on the origin of the elevated [OIII]/[CII] ratios in the early Universe. *arXiv e-prints*, arXiv:2509.16071. doi:10.48550/arXiv.2509.16071. arXiv: 2509.16071 [astro-ph.GA]

26 van Leeuwen, I. F., Bouwens, R. J., Hodge, J. A., van der Werf, P. P., Algera, H. S. B., Schouws, S., … Stefanon, M. (2025). The ALMA REBELS survey: [O III]$_{88\mu m}$ line scans of UV-bright z > 7.6 galaxies. *MNRAS*, *542*(2), 1388–1402. doi:10.1093/mnras/staf1287. arXiv: 2508.02390 [astro-ph.GA]

27 Vallini, L., Pallottini, A., Kohandel, M., **Sommovigo**, **L.**, Ferrara, A., Bethermin, M., … Accard, C. (2025). Spatially resolved [CII]–gas conversion factor in early galaxies. *A&A*, *700*, A117. doi:10.1051/0004-6361/202555179. arXiv: 2504.14001 [astro-ph.GA]

28 Pallottini, A., Ferrara, A., Gallerani, S., **Sommovigo**, **L.**, Carniani, S., Vallini, L., … Venturi, G. (2025). The mass-metallicity relation as a ruler for galaxy evolution: Insights from the James Webb Space Telescope. *A&A*, *699*, A6. doi:10.1051/0004-6361/202451742. arXiv: 2408.00061 [astro-ph.GA]

29 Schouws, S., Bouwens, R. J., Ormerod, K., Smit, R., Algera, H., **Sommovigo**, **L.**, … van der Werf, P. (2025). Detection of [O III]$_{88\mu m}$ in JADES-GS-z14-0 at z = 14.1793. *ApJ*, *988*(1), 19. doi:10.3847/1538-4357/adbf1b. arXiv: 2409.20549 [astro-ph.GA]

30 Fisher, R., Bowler, R. A. A., Stefanon, M., Rowland, L. E., Algera, H. S. B., Aravena, M., … van der Werf, P. P. (2025). REBELS-IFU: dust attenuation curves of 12 massive galaxies at z $\sim$ 7. *MNRAS*, *539*(1), 109–126. doi:10.1093/mnras/staf485. arXiv: 2501.10541 [astro-ph.GA]

31 Di Mascia, F., Pallottini, A., **Sommovigo**, **L.**, & Decataldo, D. (2025). Investigating ultraviolet and infrared radiation through the turbulent life of molecular clouds. *A&A*, *695*, A77. doi:10.1051/0004-6361/202451430. arXiv: 2407.01662 [astro-ph.GA]

32 Markov, V., Gallerani, S., Ferrara, A., Pallottini, A., Parlanti, E., Mascia, F. D., … Kohandel, M. (2025). The evolution of dust attenuation in z $\approx$ 2-12 galaxies observed by JWST. *Nature Astronomy*, *9*, 458–468. doi:10.1038/s41550-024-02426-1. arXiv: 2402.05996 [astro-ph.GA]

33 Ferrara, A., Pallottini, A., & **Sommovigo**, **L.** (2025). Blue monsters at z > 10: Where all their dust has gone. *A&A*, *694*, A286. doi:10.1051/0004-6361/202452707. arXiv: 2410.19042 [astro-ph.GA]

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Conferences & talks (selected)

Invited Talks

2026 **MIAPbP**, "Continuing the JWST Revolution: Understanding Early Galaxy Formation", Munich, Germany

2026 **"The Drivers of Galaxy Evolution through Cosmic Time"**, Florence, Italy

2026 **"Charting Cosmic Dawn in Copenhagen 2026"**, Copenhagen, Denmark

2025 **2025 Simons Collaboration on Learning the Universe Annual Meeting**, New York, USA

2025 **Kaba-Kada Conference**, "Exploring the first billion years of the Universe", Queensland, Australia

2024 **KITP**, "Cosmic Origins: The First Billion Years", Santa Barbara, USA

2024 **IAU General Assembly**, IAUS-391, Cape Town, South Africa

2024 **EAS**, SS09, Padova, Italy

2024 **Lorentz Center Workshop**, "Synergistic ALMA+JWST view of the early universe", Leiden, Netherlands

2024 **Cosmic Odysseys 2024**, Crete, Greece

2022 **Princeton University Star Formation/ISM Rendezvous (SFIR)**, The Institute for Advanced Study School of Natural Sciences, Princeton, NJ 08540 USA

2022 **Carnegie Tea Talk**, Carnegie Observatories, Pasadena, CA 91101 USA

2022 **Lorentz Center Workshop**, "Mapping the Invisible Universe", Leiden, Netherlands

2022 **Monday lunch talk**, University of Groningen

2022 **"Linking galaxy physics from ISM to IGM scales" Conference**, Sexten, Italy

Conferences & talks (selected) (continued)

2020 **Galaxy Crawl virtual talks**, Arizona, United States

Contributed Talks

2024 **Evolution of Dust and Gas throughout Cosmic Time (DG2024)**, Hiroshima, Japan
2024 **Cosmic Origins (firstbillion24)**, KITP program, Santa Barbara, USA
2024 **First Stars VII**, New York, USA
2023 **Resolving the Extragalactic Universe with ALMA & JWST**, Tokyo, Japan
2023 **Fifth Workshop on Millimetre Astronomy in Italy**, Italy
2022 **Charting the metallicity evolution history of the Universe**, Catania, Italy
2022 **From galaxies to cosmology with large spectroscopic surveys**, Marseille, France
2022 **IR STIG: Infra-Red Science and Technology Integration Group**, Colorado Boulder, USA
2022 **SAZERAC SIP: Metals and dust at high redshift**, Online conference
2022 **Quasars and galaxies through cosmic time**, Online conference
2021 **SAZERAC SIP: Models and Simulations of High-Redshift Galaxies**, Online conference
2021 **EWASS**, European Week of Astronomy and Space Science, Online conference
2021 **SAZERAC CIDER: The Cold ISM During the Epoch of Reionization**, Online conference
2020 **The rise of metals and dust through cosmic time**, Hybrid Conference, Marseille, France
2020 **SAZERAC: the Summer All-Zoom Epoch of Reionization Astronomy Conference**, Online conference
2020 **The Interstellar medium of high redshift galaxies**, Sexten, Italy
2019 **Zoom-In and Out: From the Interstellar Medium to the Large Scale Structure of the Universe**, Nordita, Stockholm, Sweden

Seminars & Colloquia

2024 **CTC Seminar**, University of Maryland, College Park, Maryland
2022 **ESO Hypatia Colloquium**, Online colloquium
2022 **Galaxy formation seminar**, Oxford, United Kingdom
2022 **Remote Seminar at MPE**, Garching, Germany
2021 **UniMelb Astro Colloquium**, Melbourne, Australia (online)

Skills

Languages	◇ Council of Europe level C2 in English, Native Italian speaker
Coding	◇ Python, C, LaTeX
Misc.	◇ Academic research, co-supervision of students, design and coordination of projects aimed at increasing inclusivity and diversity in academia, experience in scientific and outreach talks